Project Style Guide and Glossary

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1 Style Guide & Glossary

1.1 Purpose

This document defines naming, formatting, and style conventions for reports, scripts, metadata, and documentation used in the project. Additionally, the document acts as a dictionary of project-specific vocabulary.

1.2 Terminology and Naming Conventions

1.2.1 Macrozone Names

Macrozone	Preferred Term	Notes
`tallgrass`	Tallgrass Prairie	No hyphen; conventional in literature
`mixed`	Mixed-Grass Prairie	Hyphenated compound modifier
`shortgrass`	Shortgrass-Steppe	Hyphenated; “Prairie” is redundant

Use hyphenated forms when modifying a noun (e.g., “Shortgrass-Steppe region”), and unhyphenated when used as a noun (e.g., “We sampled the Shortgrass Steppe”).

1.3 Variable Naming

1.3.1 General Use Cases

- Use snake_case for all variable names (e.g., `ppt_ann_mm`, `slope_pct`)
- Place `site_no`, `latitude`, and `longitude` as the first columns in exported datasets
- Use standard units in suffixes (`_mm`, `_C`, `_pct`, `_km2`)

1.3.2 Specific Use Cases for Spatial Data Workflows

The project strives to use explicitly named, project-specific ID fields, which are treated as a UID. One qc check on asserting UID status prior to a join is by using the `assert_inputs_ok()` helper function. The reason UIDs are important is to ensure one to one cardinality in joining or merging datasets.

- **site_no** — USGS gage ID (UID, not regenerated).
- **macro_id** — persistent UID for macrozones (sequential, unique).
- **catch_id** — NHDPlus catchment UID.

1.4 Writing Style

- Use full phrases before acronyms on first use (e.g., "Normalized Difference Vegetation Index (NDVI)").
- Follow USGS and EPA naming conventions when describing datasets, flowlines, and catchments.
- Avoid unnecessary repetition, in other words tautologies. For example, avoid writing "Shortgrass-Steppe Prairie".

1.5 Citation Style

- Internal datasets: reference relative path in `docs/metadata/`.
- External datasets: provide DOI or citation in README/data dictionary.

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2 Appendix

2.1 An Explaination of ID-term in Spatial Data Workflows

- **ID (Identifier)** is the *generic term* for a field that uniquely (or semi-uniquely) identifies records, and often consists of sequential integers assigned on export. While it is commonly used in shapefiles/GeoPackages (ID, id, FID), it *may not be stable across exports or joins*.

- **UID (Unique Identifier)** is a term that explicitly means *guaranteed unique* within the dataset. A UID provides a stronger guarantee than plain ID, meaning no duplicates and no recycling. A UID is often in a UUID/GUID format (e.g., 550e8400-e29b-41d4-a716-446655440000) or carefully managed integer sequences. UID is *useful when performing merge/join across multiple datasets*, and when persistence across time and versions is needed

- **OID (Object ID)** is a term from Esri geodatabases.and is an automatically maintained field (OBJECTID or OID). OID is internal to the geodatabase and ensures every feature has a unique row identifier. *OID is* **not** *stable upon re-export or recompute*, as Esri may regenerate an OID. Therefore, it should not be used as an analysis join key in persistant workflows.